$$
\begin{aligned}
&C_{e\phi_2}^{(11)}{}_{i1\_i2\_} \to \frac{1}{16\pi^2}\Big(\tfrac{1}{4} g_1^4 y_e^{i1i2} LF_{2,2,-1}[m_1, m_{\tilde e}^{i2}] - \\
&\tfrac{1}{32} g_1^2 y_e^{i1i2}(g_1^2+g_2^2) LF_{2,2,-1}[m_1, m_{\tilde l}^{i1}] + \tfrac{1}{32} g_2^2 y_e^{i1i2}(-3g_1^2+g_2^2) LF_{2,2,-1}[m_2, m_{\tilde l}^{i1}] + \\
&\tfrac14 g_1^2 y_e^{i1i2}\overline{a_d}^{pr} a_d^{pr} LF_{2,2,0}[m_{\tilde d}^r, m_{\tilde q}^p] + \tfrac14 g_1^2 y_e^{i1i2}\overline{a_e}^{pr} a_e^{pr} LF_{2,2,0}[m_{\tilde e}^r, m_{\tilde l}^p] - \\
&\tfrac14 g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,0}[m_{\tilde e}^r, \tilde\mu] + \tfrac18 g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,2,-1}[m_{\tilde e}^r, \tilde\mu] - \\
&\tfrac12 g_1^4 y_e^{i1i2} LF_{2,1,0}[m_{\tilde e}^{i2}, m_1] - \tfrac12 g_2^2 y_e^{i1i2}\overline{a_e}^{pr} a_e^{pr} LF_{3,1,0}[m_{\tilde l}^p, m_{\tilde e}^r] - \\
&\tfrac14 g_1^2 y_e^{i1i2}\overline{a_e}^{pr} a_e^{pr} LF_{3,2,-1}[m_{\tilde l}^p, m_{\tilde e}^r] - \tfrac38 y_e^{i1i2}\overline{a_e}^{pr} a_e^{pr}(g_1^2-3g_2^2) LF_{4,1,-1}[m_{\tilde l}^p, m_{\tilde e}^r] + \\
&\tfrac12 y_e^{i1i2}\overline{a_e}^{pr} a_e^{pr}(g_1^2-g_2^2) LF_{5,1,-2}[m_{\tilde l}^p, m_{\tilde e}^r] + \\
&\tfrac14 g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,0}[m_{\tilde l}^p, \tilde\mu] - \tfrac18 g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,2,-1}[m_{\tilde l}^p, \tilde\mu] + \\
&\tfrac{1}{16} g_1^2 y_e^{i1i2}(g_1^2+g_2^2) LF_{2,1,0}[m_{\tilde l}^{i1}, m_1] - \tfrac{1}{16} g_2^2 y_e^{i1i2}(-3g_1^2+g_2^2) LF_{2,1,0}[m_{\tilde l}^{i1}, m_2] + \\
&\tfrac12 y_e^{i1i2}\overline{a_d}^{pr} a_d^{pr}(2g_1^2-3g_2^2) LF_{3,1,0}[m_{\tilde q}^p, m_{\tilde d}^r] - \tfrac14 g_1^2 y_e^{i1i2}\overline{a_d}^{pr} a_d^{pr} LF_{3,2,-1}[m_{\tilde q}^p, m_{\tilde d}^r] - \\
&\tfrac38 y_e^{i1i2}\overline{a_d}^{pr} a_d^{pr}(7g_1^2-9g_2^2) LF_{4,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r] + \\
&\tfrac32 y_e^{i1i2}\overline{a_d}^{pr} a_d^{pr}(g_1^2-g_2^2) LF_{5,1,-2}[m_{\tilde q}^p, m_{\tilde d}^r] + \tfrac18 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{pr} y_u^{pr}(g_1^2+3g_2^2) \\
&\quad LF_{2,2,0}[m_{\tilde q}^p, m_{\tilde u}^r] - \tfrac14 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{pr} y_u^{pr}(g_1^2+3g_2^2) LF_{3,1,0}[m_{\tilde u}^r, m_{\tilde q}^p] - \\
&\tfrac18 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{pr} y_u^{pr}(g_1^2+3g_2^2) LF_{3,2,-1}[m_{\tilde u}^r, m_{\tilde q}^p] - \tfrac34 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{pr} y_u^{pr} \\
&\quad (g_1^2-3g_2^2) LF_{4,1,-1}[m_{\tilde u}^r, m_{\tilde q}^p] + \tfrac32 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{pr} y_u^{pr}(g_1^2-g_2^2) LF_{5,1,-2}[m_{\tilde u}^r, m_{\tilde q}^p] + \\
&\tfrac32 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2}\overline{y_u}^{st} y_u^{pt} LF_{2,1,0}[m_{\tilde u}^t, m_{\tilde d}^r] - \tfrac32 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2}\overline{y_u}^{st} y_u^{pt} LF_{3,1,-1}[m_{\tilde u}^t, m_{\tilde d}^r] + \\
&\tfrac12 g_1^2 y_e^{i1i2}(-g_1^2+g_2^2) LF_{3,1,-1}[\tilde\mu, m_1] - \tfrac38 g_1^4 y_e^{i1i2} LF_{3,2,-2}[\tilde\mu, m_1] - \\
&\tfrac38 g_1^4 m_1^2 y_e^{i1i2} LF_{3,2,-1}[\tilde\mu, m_1] + g_1^2 y_e^{i1i2}(g_1^2-g_2^2) LF_{4,1,-2}[\tilde\mu, m_1] + \\
&\tfrac14 g_1^4 y_e^{i1i2} LF_{4,2,-3}[\tilde\mu, m_1] + \tfrac14 g_1^4 m_1^2 y_e^{i1i2} LF_{4,2,-2}[\tilde\mu, m_1] + \\
&\tfrac12 g_1^2 y_e^{i1i2}(-g_1^2+g_2^2) LF_{5,1,-3}[\tilde\mu, m_1] + \tfrac32 g_2^2 y_e^{i1i2}(-g_1^2+g_2^2) LF_{3,1,-1}[\tilde\mu, m_2] - \\
&\tfrac{15}{8} g_2^4 y_e^{i1i2} LF_{3,2,-2}[\tilde\mu, m_2] - \tfrac38 g_2^4 m_2^2 y_e^{i1i2} LF_{3,2,-1}[\tilde\mu, m_2] + \\
&3 g_2^2 y_e^{i1i2}(g_1^2-g_2^2) LF_{4,1,-2}[\tilde\mu, m_2] + \tfrac54 g_2^4 y_e^{i1i2} LF_{4,2,-3}[\tilde\mu, m_2] + \\
&\tfrac14 g_2^4 y_e^{i1i2}(m_2^2+2\tilde\mu^2) LF_{4,2,-2}[\tilde\mu, m_2] + \tfrac32 g_2^2 y_e^{i1i2}(-g_1^2+g_2^2) LF_{5,1,-3}[\tilde\mu, m_2] + \\
&\tfrac{1}{16} m_1 g_1^2 a_e^{i1i2}(-g_1^2+g_2^2) LF_{2,2,1,-1}[m_1, m_{\tilde e}^{i2}, m_{\tilde l}^{i1}] - \\
&\tfrac34 g_1^4 y_e^{i1i2} LF_{2,1,1,-1}[m_1, m_{\tilde e}^{i2}, \tilde\mu] - \tfrac12 g_1^4 m_1^2 y_e^{i1i2} LF_{2,1,1,0}[m_1, m_{\tilde e}^{i2}, \tilde\mu] + \\
&\tfrac12 g_1^4 y_e^{i1i2} LF_{3,1,1,-2}[m_1, m_{\tilde e}^{i2}, \tilde\mu] + \tfrac12 g_1^4 m_1^2 y_e^{i1i2} LF_{3,1,1,-1}[m_1, m_{\tilde e}^{i2}, \tilde\mu] + \\
&\tfrac{1}{16} m_1 g_1^2 a_e^{i1i2}(-g_1^2+g_2^2) LF_{2,2,1,-1}[m_1, m_{\tilde l}^{i1}, m_{\tilde e}^{i2}] - \tfrac{3}{16} g_1^4 y_e^{i1i2} LF_{2,1,1,-1}[m_1, m_{\tilde l}^{i1}, \tilde\mu] - \\
&\tfrac18 g_1^4 m_1^2 y_e^{i1i2} LF_{2,1,1,0}[m_1, m_{\tilde l}^{i1}, \tilde\mu] + \tfrac18 g_1^4 y_e^{i1i2} LF_{3,1,1,-2}[m_1, m_{\tilde l}^{i1}, \tilde\mu] + \\
&\tfrac18 g_1^4 m_1^2 y_e^{i1i2} LF_{3,1,1,-1}[m_1, m_{\tilde l}^{i1}, \tilde\mu] + \tfrac18 g_1^2 y_e^{i1i2}(7g_1^2-g_2^2) LF_{2,2,1,-2}[m_1, \tilde\mu, m_{\tilde e}^{i2}] + \\
&\tfrac34 g_1^4 m_1^2 y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde e}^{i2}] + \tfrac18 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde l}^p] + \\
&\tfrac{1}{16} g_1^2 y_e^{i1i2}(-4g_1^2+g_2^2) LF_{2,2,1,-2}[m_1, \tilde\mu, m_{\tilde l}^{i1}] - \tfrac{3}{16} g_1^4 m_1^2 y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde l}^{i1}] - \\
&\tfrac{15}{16} g_2^4 y_e^{i1i2} LF_{2,1,1,-1}[m_2, m_{\tilde l}^{i1}, \tilde\mu] - \tfrac18 g_2^4 m_2^2 y_e^{i1i2} LF_{2,1,1,0}[m_2, m_{\tilde l}^{i1}, \tilde\mu] + \\
&\tfrac58 g_2^4 y_e^{i1i2} LF_{3,1,1,-2}[m_2, m_{\tilde l}^{i1}, \tilde\mu] + \tfrac18 g_2^4 m_2^2 y_e^{i1i2} LF_{3,1,1,-1}[m_2, m_{\tilde l}^{i1}, \tilde\mu] - \\
&3 g_1^2 g_2^2 \tilde\mu^2 y_e^{i1i2} LF_{2,2,1,-1}[m_2, \tilde\mu, m_1] + 2 g_1^2 g_2^2 \tilde\mu^2 y_e^{i1i2} LF_{3,2,1,-2}[m_2, \tilde\mu, m_1] + \\
&\tfrac14 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde e}^r] + \tfrac38 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde l}^p] + \\
&\tfrac{1}{16} g_2^2 y_e^{i1i2}(3g_1^2+22g_2^2) LF_{2,2,1,-2}[m_2, \tilde\mu, m_{\tilde l}^{i1}] + \tfrac{5}{16} g_2^4 m_2^2 y_e^{i1i2} LF_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde l}^{i1}] + \\
&\tfrac18 m_1 g_1^2 a_e^{i1i2}(5g_1^2-g_2^2) LF_{2,1,1,0}[m_{\tilde e}^{i2}, m_1, m_{\tilde l}^{i1}] + \\
&\tfrac18 m_1 g_1^2 a_e^{i1i2}(-g_1^2+g_2^2) LF_{3,1,1,-1}[m_{\tilde e}^{i2}, m_1, m_{\tilde l}^{i1}] + \\
&\tfrac12 g_1^4 y_e^{i1i2} LF_{2,1,1,-1}[m_{\tilde e}^{i2}, m_1, \tilde\mu] - \tfrac18 m_1 g_1^2 a_e^{i1i2}(g_1^2+3g_2^2) LF_{2,1,1,0}[m_{\tilde l}^{i1}, m_1, m_{\tilde e}^{i2}] + \\
&\tfrac18 m_1 g_1^2 a_e^{i1i2}(-g_1^2+g_2^2) LF_{3,1,1,-1}[m_{\tilde l}^{i1}, m_1, m_{\tilde e}^{i2}] + \tfrac18 g_1^2 y_e^{i1i2}(g_1^2+g_2^2) \\
&\quad LF_{2,1,1,-1}[m_{\tilde l}^{i1}, m_1, \tilde\mu] + \tfrac18 g_2^2 y_e^{i1i2}(-3g_1^2+g_2^2) LF_{2,1,1,-1}[m_{\tilde l}^{i1}, m_2, \tilde\mu] + \\
&3 y_e^{i1i2}\overline{y_u}^{st} y_u^{pt}\overline{a_d}^{pr} a_d^{sr} LF_{2,1,1,0}[m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde q}^s] - 3 y_e^{i1i2}\overline{y_u}^{st} y_u^{pt}\overline{a_d}^{pr} a_d^{sr} \\
&\quad LF_{3,1,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde q}^s] + 3 y_d^{sr} y_e^{i1i2}\overline{y_u}^{st}\overline{a_d}^{pr} a_u^{pt} LF_{2,1,1,0}[m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde u}^t] - \\
&3 y_d^{sr} y_e^{i1i2}\overline{y_u}^{st}\overline{a_d}^{pr} a_u^{pt} LF_{3,1,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde u}^t] - \tfrac32 y_e^{i1i2}\overline{y_u}^{st} y_u^{pt}\overline{a_d}^{pr} a_d^{sr} \\
&\quad LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r] - \tfrac34 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt}\overline{a_u}^{pr} LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde d}^t] - \\
&\tfrac34 y_d^{sr} y_e^{i1i2}\overline{y_u}^{st}(3\overline{a_d}^{pr} a_u^{pt}+\tilde\mu^2\overline{y_d}^{pr} y_u^{pt}) LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde d}^r] + \\
&\tfrac34 \tilde\mu^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2}\overline{y_u}^{st} y_u^{pt} LF_{2,2,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde d}^r] + \\
&\tfrac32 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt}\overline{a_u}^{pr} LF_{2,1,1,0}[m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p] - \\
&\tfrac32 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt}\overline{a_u}^{pr} LF_{3,1,1,-1}[m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p] + \\
&3 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{pt} y_u^{sr} LF_{2,1,1,0}[m_{\tilde u}^r, m_{\tilde q}^s, m_{\tilde u}^t] - 3 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{pt} y_u^{sr} \\
&\quad LF_{3,1,1,-1}[m_{\tilde u}^r, m_{\tilde q}^s, m_{\tilde u}^t] - \tfrac32 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{pr}\overline{y_u}^{st} y_u^{pt} y_u^{sr} LF_{2,2,1,-1}[m_{\tilde u}^r, m_{\tilde u}^t, m_{\tilde q}^s] + \\
&\tfrac32 y_d^{sr} y_e^{i1i2}\overline{y_u}^{st}(\overline{a_d}^{pr} a_u^{pt}-\tilde\mu^2\overline{y_d}^{pr} y_u^{pt}) LF_{2,1,1,0}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p] + \\
&\tfrac32 y_d^{sr} y_e^{i1i2}\overline{y_u}^{st}(-\overline{a_d}^{pr} a_u^{pt}+\tilde\mu^2\overline{y_d}^{pr} y_u^{pt}) LF_{3,1,1,-1}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p] - \\
&\tfrac32 \tilde\mu^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2}\overline{y_u}^{st} y_u^{pt} LF_{2,1,1,0}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^s] + \\
&\tfrac32 \tilde\mu^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2}\overline{y_u}^{st} y_u^{pt} LF_{3,1,1,-1}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^s] + \\
&\tfrac34 g_1^2 g_2^2 y_e^{i1i2} LF_{3,1,1,-2}[\tilde\mu, m_1, m_2] + \tfrac34 g_1^2 g_2^2 y_e^{i1i2}(m_1 m_2-4\tilde\mu^2) LF_{3,1,1,-1}[\tilde\mu, m_1, m_2] - \\
&\tfrac12 g_1^2 g_2^2 y_e^{i1i2} LF_{4,1,1,-3}[\tilde\mu, m_1, m_2] - \tfrac12 g_1^2 g_2^2 y_e^{i1i2}(m_1 m_2-5\tilde\mu^2) LF_{4,1,1,-2}[\tilde\mu, m_1, m_2] - \\
&\tfrac14 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde l}^p] + \tfrac14 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde l}^p] + \\
&2 g_1^2 g_2^2 \tilde\mu^2 y_e^{i1i2} LF_{3,2,1,-2}[\tilde\mu, m_2, m_1] - \tfrac12 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde e}^r] + \\
&\tfrac12 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde e}^r] - \tfrac34 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde l}^p] + \\
&\tfrac34 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde l}^p] - \tfrac38 g_1^2 g_2^2 y_e^{i1i2} LF_{1,1,1,1,-1}[m_1, m_2, m_{\tilde l}^{i1}, \tilde\mu] - \\
&\tfrac14 m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{1,1,1,1,0}[m_1, m_2, m_{\tilde l}^{i1}, \tilde\mu] + \tfrac18 g_1^2 g_2^2 y_e^{i1i2} \\
&\quad LF_{2,1,1,1,-2}[m_1, m_2, m_{\tilde l}^{i1}, \tilde\mu] + \tfrac18 m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-1}[m_1, m_2, m_{\tilde l}^{i1}, \tilde\mu] - \\
&\tfrac34 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde e}^r, m_{\tilde l}^p, \tilde\mu] + \tfrac38 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad LF_{2,1,1,1,-1}[m_1, m_{\tilde e}^r, m_{\tilde l}^p, \tilde\mu] - g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde e}^{i2}, m_{\tilde e}^r, m_{\tilde l}^p] + \\
&\tfrac12 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_1, m_{\tilde e}^{i2}, m_{\tilde e}^r, m_{\tilde l}^p] - \tfrac14 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad LF_{1,1,1,1,0}[m_1, m_{\tilde e}^{i2}, m_{\tilde l}^p, \tilde\mu] + \tfrac18 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_1, m_{\tilde e}^{i2}, m_{\tilde l}^p, \tilde\mu] + \\
&m_1 g_1^4 a_e^{i1i2} LF_{2,1,1,1,-1}[m_1, m_{\tilde e}^{i2}, m_{\tilde l}^{i1}, \tilde\mu] + \tfrac18 g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-2}[m_2, m_1, m_{\tilde l}^{i1}, \tilde\mu] + \\
&\tfrac18 m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-1}[m_2, m_1, m_{\tilde l}^{i1}, \tilde\mu] - \\
&\tfrac34 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde e}^r, m_{\tilde l}^p, \tilde\mu] + \\
&\tfrac38 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_2, m_{\tilde e}^r, m_{\tilde l}^p, \tilde\mu] - \\
&\tfrac12 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde e}^r, m_{\tilde l}^{i1}, m_{\tilde l}^p] + \\
&\tfrac14 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_2, m_{\tilde e}^r, m_{\tilde l}^{i1}, m_{\tilde l}^p] - \\
&\tfrac14 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde e}^r, m_{\tilde l}^{i1}, \tilde\mu] + \\
&\tfrac18 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_2, m_{\tilde e}^r, m_{\tilde l}^{i1}, \tilde\mu] + \\
&\tfrac34 \tilde\mu^2 \overline{y_d}^{st} y_d^{sr} y_e^{i1i2}\overline{a_d}^{pr} a_d^{pt} LF_{2,2,1,1,-1}[m_{\tilde d}^r, m_{\tilde q}^p, m_{\tilde d}^t, m_{\tilde q}^s] - \\
&\tfrac34 \tilde\mu^2 \overline{y_d}^{st} y_d^{sr} y_e^{i1i2}\overline{a_d}^{pr} a_d^{pt} LF_{2,2,1,1,-1}[m_{\tilde d}^t, m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde q}^s] - \\
&\tfrac12 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde e}^r, m_1, m_{\tilde e}^{i2}, m_{\tilde l}^p] - \\
&\tfrac18 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde e}^r, m_1, m_{\tilde l}^p, \tilde\mu] + \\
&\tfrac18 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde e}^r, m_2, m_{\tilde l}^p, \tilde\mu] - \\
&\tfrac18 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde e}^r, m_2, m_{\tilde l}^{i1}, \tilde\mu] - \\
&\tfrac12 \tilde\mu^2 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} LF_{1,1,1,1,0}[m_{\tilde e}^r, m_{\tilde e}^t, m_{\tilde l}^s, \tilde\mu] - \\
&\tfrac14 \tilde\mu^2 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} LF_{2,1,1,1,-1}[m_{\tilde e}^r, m_{\tilde e}^t, m_{\tilde l}^s, \tilde\mu] + \\
&\tfrac14 \tilde\mu^2 \overline{y_e}^{st} y_e^{sr} y_e^{i1i2}\overline{a_e}^{pr} a_e^{pt} LF_{2,2,1,1,-1}[m_{\tilde e}^r, m_{\tilde l}^p, m_{\tilde e}^t, m_{\tilde l}^s] + \\
&\tfrac14 \tilde\mu^2 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} LF_{2,1,1,1,-1}[m_{\tilde e}^t, m_{\tilde e}^r, m_{\tilde l}^s, \tilde\mu] - \\
&\tfrac14 \tilde\mu^2 \overline{y_e}^{st} y_e^{sr} y_e^{i1i2}\overline{a_e}^{pr} a_e^{pt} LF_{2,2,1,1,-1}[m_{\tilde e}^t, m_{\tilde l}^p, m_{\tilde e}^r, m_{\tilde l}^s] - \\
&\tfrac12 \tilde\mu^2 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{pt} y_e^{si2} y_e^{i1r} LF_{1,1,1,1,0}[m_{\tilde e}^t, m_{\tilde l}^p, m_{\tilde l}^s, \tilde\mu] + \\
&\tfrac12 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde e}^{i2}, m_1, m_{\tilde e}^r, m_{\tilde l}^p] + \\
&\tfrac18 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde e}^{i2}, m_1, m_{\tilde l}^p, \tilde\mu] + \\
&\tfrac18 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde l}^p, m_1, m_{\tilde e}^r, \tilde\mu] - \\
&\tfrac18 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde l}^p, m_1, m_{\tilde e}^{i2}, \tilde\mu] - \\
&\tfrac14 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde l}^p, m_2, m_{\tilde e}^r, m_{\tilde l}^{i1}] - \\
&\tfrac18 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde l}^p, m_2, m_{\tilde e}^r, \tilde\mu] + \\
&\tfrac12 \tilde\mu^2 \overline{y_e}^{st} y_e^{sr} y_e^{i1i2}\overline{a_e}^{pr} a_e^{pt} LF_{2,1,1,1,0}[m_{\tilde l}^p, m_{\tilde e}^r, m_{\tilde e}^t, m_{\tilde l}^s] - \\
&\tfrac12 \tilde\mu^2 \overline{y_e}^{st} y_e^{sr} y_e^{i1i2}\overline{a_e}^{pr} a_e^{pt} LF_{3,1,1,1,-1}[m_{\tilde l}^p, m_{\tilde e}^r, m_{\tilde e}^t, m_{\tilde l}^s] + \\
&\tfrac14 \tilde\mu^2 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} LF_{2,1,1,1,-1}[m_{\tilde l}^p, m_{\tilde e}^r, m_{\tilde l}^s, \tilde\mu] - \\
&\tfrac14 \tilde\mu^2 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{pt} y_e^{si2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde l}^p, m_{\tilde e}^t, m_{\tilde l}^s, \tilde\mu] + \\
&\tfrac14 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde l}^{i1}, m_2, m_{\tilde e}^r, m_{\tilde l}^p] + \\
&\tfrac18 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde l}^{i1}, m_2, m_{\tilde e}^r, \tilde\mu] + \\
&\tfrac32 \tilde\mu^2 \overline{y_d}^{st} y_d^{sr} y_e^{i1i2}\overline{a_d}^{pr} a_d^{pt} LF_{2,1,1,1,0}[m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde d}^t, m_{\tilde q}^s] - \\
&\tfrac32 \tilde\mu^2 \overline{y_d}^{st} y_d^{sr} y_e^{i1i2}\overline{a_d}^{pr} a_d^{pt} LF_{3,1,1,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde d}^t, m_{\tilde q}^s] + \\
&3 y_e^{i1i2}\overline{a_d}^{pr} a_u^{pt}(a_d^{sr}\overline{a_u}^{st}-\tilde\mu^2 y_d^{sr}\overline{y_u}^{st}) LF_{2,1,1,1,0}[m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde q}^s, m_{\tilde u}^t] + \\
&3 y_e^{i1i2}\overline{a_d}^{pr} a_u^{pt}(-a_d^{sr}\overline{a_u}^{st}+\tilde\mu^2 y_d^{sr}\overline{y_u}^{st}) LF_{3,1,1,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r, m_{\tilde q}^s, m_{\tilde u}^t] + \\
&\tfrac34 y_e^{i1i2}\overline{a_d}^{pr} a_u^{pt}(-2 a_d^{sr}\overline{a_u}^{st}+\tilde\mu^2 y_d^{sr}\overline{y_u}^{st}) LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r, m_{\tilde u}^t] - \\
&\tfrac34 \tilde\mu^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt}\overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^t, m_{\tilde u}^r] + \\
&\tfrac34 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{2,1,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] - \\
&\tfrac34 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{3,1,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] + \\
&\tfrac34 \tilde\mu^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt}\overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^s] - \\
&\tfrac98 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde q}^s, m_{\tilde u}^t] + \\
&\tfrac34 y_e^{i1i2}(\tilde\mu^4 \overline{y_d}^{pr} y_d^{sr}\overline{y_u}^{st} y_u^{pt}+3\overline{a_d}^{pr} a_u^{pt}(-a_d^{sr}\overline{a_u}^{st}+\tilde\mu^2 y_d^{sr}\overline{y_u}^{st})) LF_{2,2,1,1,-1}[m_{\tilde q}^p, \\
&\quad m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^s] - \tfrac38 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde q}^s, m_{\tilde u}^r] + \\
&\tfrac34 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{2,1,1,1,0}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde u}^t] - \\
&\tfrac34 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{3,1,1,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde u}^t] - \\
&\tfrac34 \tilde\mu^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt}\overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p] + \\
&\tfrac38 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde q}^p, m_{\tilde u}^t] - \tfrac34 y_e^{i1i2} \\
&\quad (\tilde\mu^4 \overline{y_d}^{pr} y_d^{sr}\overline{y_u}^{st} y_u^{pt}+\overline{a_d}^{pr} a_u^{pt}(a_d^{sr}\overline{a_u}^{st}-\tilde\mu^2 y_d^{sr}\overline{y_u}^{st})) LF_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p] - \\
&\tfrac38 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde q}^p, m_{\tilde u}^r] - \\
&\tfrac32 \tilde\mu^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt}\overline{a_u}^{pr} LF_{2,1,1,1,0}[m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p, m_{\tilde q}^s] + \\
&\tfrac32 \tilde\mu^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt}\overline{a_u}^{pr} LF_{3,1,1,1,-1}[m_{\tilde u}^r, m_{\tilde d}^t, m_{\tilde q}^p, m_{\tilde q}^s] + \\
&\tfrac32 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{2,1,1,1,0}[m_{\tilde u}^r, m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^t] - \\
&\tfrac32 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{3,1,1,1,-1}[m_{\tilde u}^r, m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^t] - \\
&\tfrac34 \tilde\mu^2 y_e^{i1i2}\overline{y_u}^{st} y_u^{sr}\overline{a_u}^{pr} a_u^{pt} LF_{2,2,1,1,-1}[m_{\tilde u}^r, m_{\tilde u}^t, m_{\tilde q}^p, m_{\tilde q}^s] + \tfrac32 y_e^{i1i2} \\
&\quad (\tilde\mu^4 \overline{y_d}^{pr} y_d^{sr}\overline{y_u}^{st} y_u^{pt}+\overline{a_d}^{pr} a_u^{pt}(a_d^{sr}\overline{a_u}^{st}-\tilde\mu^2 y_d^{sr}\overline{y_u}^{st})) LF_{2,1,1,1,0}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p, m_{\tilde q}^s] - \\
&\tfrac32 y_e^{i1i2}(\tilde\mu^4 \overline{y_d}^{pr} y_d^{sr}\overline{y_u}^{st} y_u^{pt}+\overline{a_d}^{pr} a_u^{pt}(a_d^{sr}\overline{a_u}^{st}-\tilde\mu^2 y_d^{sr}\overline{y_u}^{st})) \\
&\quad LF_{3,1,1,1,-1}[m_{\tilde u}^t, m_{\tilde d}^r, m_{\tilde q}^p, m_{\tilde q}^s] - \tfrac12 g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-2}[\tilde\mu, m_1, m_2, m_{\tilde e}^{i2}] - \\
&\tfrac12 g_1^2 g_2^2 y_e^{i1i2}(m_1 m_2-4\tilde\mu^2) LF_{2,1,1,1,-1}[\tilde\mu, m_1, m_2, m_{\tilde e}^{i2}] + \\
&\tfrac18 g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-2}[\tilde\mu, m_1, m_2, m_{\tilde l}^{i1}] + \tfrac18 m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \\
&\quad LF_{2,1,1,1,-1}[\tilde\mu, m_1, m_2, m_{\tilde l}^{i1}] + \tfrac58 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde e}^r, m_{\tilde l}^p] + \\
&\tfrac38 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde e}^{i2}, m_{\tilde l}^p] + \tfrac58 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad LF_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde e}^r, m_{\tilde l}^p] + \tfrac38 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde e}^r, m_{\tilde l}^{i1}] + \\
&\tfrac14 \tilde\mu^2 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} LF_{2,1,1,1,-1}[\tilde\mu, m_{\tilde e}^r, m_{\tilde e}^t, m_{\tilde l}^s] + \\
&\tfrac14 \tilde\mu^2 \overline{y_e}^{pr}\overline{y_e}^{st} y_e^{pt} y_e^{si2} y_e^{i1r} LF_{2,1,1,1,-1}[\tilde\mu, m_{\tilde e}^t, m_{\tilde l}^p, m_{\tilde l}^s] + \\
&\tfrac12 g_1^2 g_2^2(a_e^{i1i2}(m_1+m_2)+4\tilde\mu^2 y_e^{i1i2}) LF_{1,1,1,1,1,-1}[m_1, m_2, m_{\tilde e}^{i2}, m_{\tilde l}^{i1}, \tilde\mu] + \\
&\tfrac12 m_1 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} LF_{1,1,1,1,1,0}[m_1, m_{\tilde e}^r, m_{\tilde l}^{i1}, m_{\tilde l}^p, \tilde\mu] + \\
&g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,1,-1}[m_1, m_{\tilde e}^{i2}, m_{\tilde e}^r, m_{\tilde l}^p, \tilde\mu] + \\
&m_1 g_1^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} LF_{1,1,1,1,1,0}[m_1, m_{\tilde e}^{i2}, m_{\tilde e}^r, m_{\tilde l}^{i1}, m_{\tilde l}^p] + \\
&g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,1,-1}[m_2, m_{\tilde e}^r, m_{\tilde l}^{i1}, m_{\tilde l}^p, \tilde\mu] + \\
&\tfrac12 m_2 g_2^2 \tilde\mu^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} LF_{1,1,1,1,1,0}[m_2, m_{\tilde e}^r, m_{\tilde l}^{i1}, m_{\tilde l}^p, \tilde\mu]\Big)
\end{aligned}
$$